

Types of Authentication

◆ Introduction:

Authentication is the process of verifying the **identity of a user or system** before granting access. It ensures that only authorized users can use the system.

◆ Types of Authentication:

1. **Password-Based Authentication**
 - Uses username and password for verification
 2. **Token-Based Authentication**
 - Uses devices or tokens (OTP, smart cards)
 3. **Biometric Authentication**
 - Uses physical characteristics like fingerprint, face
 4. **Certificate-Based Authentication**
 - Uses digital certificates for identity verification
 5. **Multi-Factor Authentication (MFA)**
 - Combines two or more authentication methods
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◆ Explain Any One (Biometric Authentication):

Definition:

Biometric authentication is a method of verifying identity using **unique biological characteristics** of a person.

Types:

- Fingerprint recognition
 - Face recognition
 - Iris/retina scan
 - Voice recognition
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Working:

1. User provides biometric data (e.g., fingerprint)
 2. System captures and converts it into digital form
 3. It compares with stored data in database
 4. If matched → access granted
 5. If not matched → access denied
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Advantages:

- High security
 - Unique for each person
 - No need to remember passwords
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Disadvantages:

- Costly hardware required
- Privacy concerns
- Possibility of errors